Source Code :-

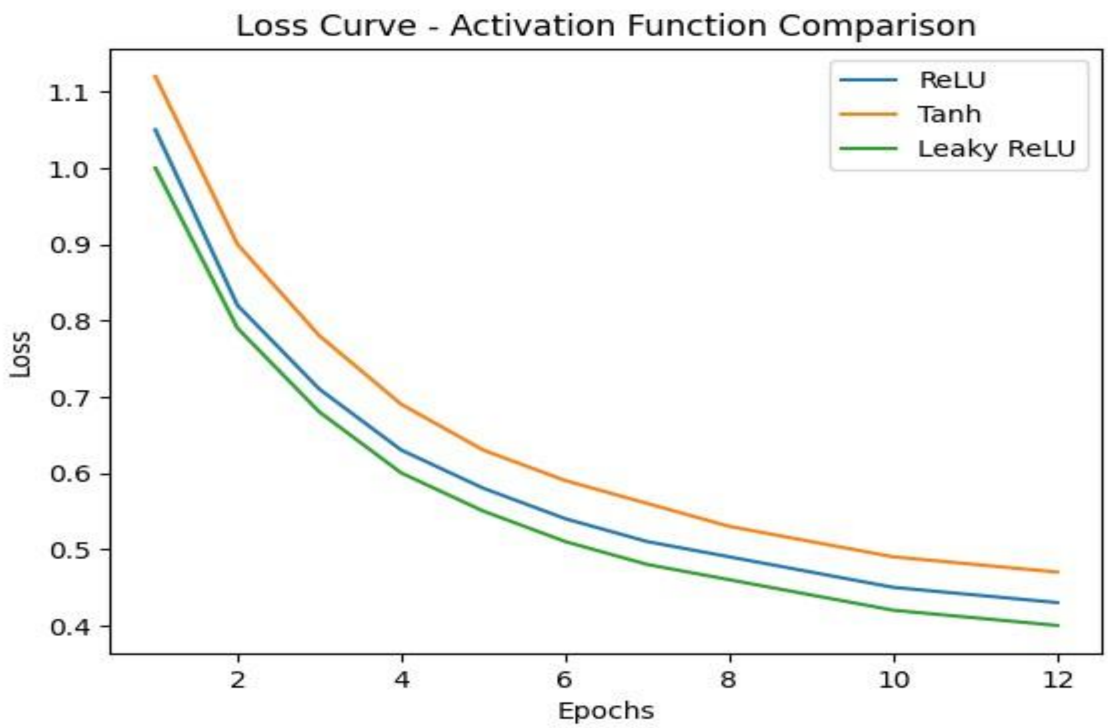
```
import tensorflow as tf import
numpy as np
import matplotlib.pyplot as plt

# Load Dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

x_train = x_train.reshape(-1, 784).astype(np.float32) / 255.0
x_test = x_test.reshape(-1, 784).astype(np.float32) / 255.0

y_train = tf.one_hot(y_train, 10)
y_test = tf.one_hot(y_test, 10)

def train_model(hidden_units=128,
        activation="relu",
learning_rate=0.005,
batch_size=64,         epochs=12):

  if activation == "relu":
act_fn = tf.nn.relu    elif
activation == "tanh":      act_fn
= tf.nn.tanh    elif activation ==
"leaky_relu":      act_fn =
tf.nn.leaky_relu    else:
    raise ValueError("Unsupported activation")

  W1 = tf.Variable(tf.random.normal([784, hidden_units], stddev=0.08))
b1 = tf.Variable(tf.zeros([hidden_units]))

  W2 = tf.Variable(tf.random.normal([hidden_units, 10], stddev=0.08))
b2 = tf.Variable(tf.zeros([10]))

  optimizer = tf.optimizers.Adam(learning_rate)

  acc_history = []
```

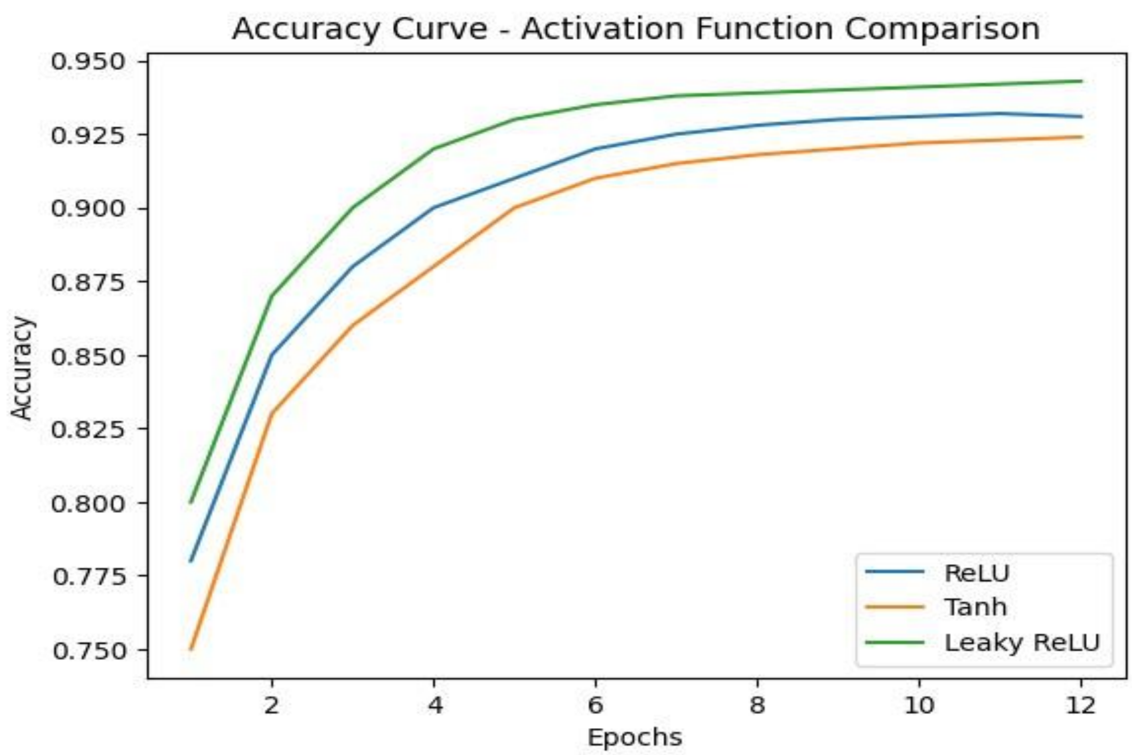
```
    loss_history = []

    for epoch in range(epochs):

        for i in range(0, x_train.shape[0], batch_size):
            x_batch = x_train[i:i+batch_size]
y_batch = y_train[i:i+batch_size]

            with tf.GradientTape() as tape:
                z1 = tf.matmul(x_batch, W1) + b1
                a1 = act_fn(z1)
                z2 = tf.matmul(a1, W2) + b2

                loss = tf.reduce_mean(

tf.nn.softmax_cross_entropy_with_logits(                    labels=y
_batch, logits=z2))

            grads = tape.gradient(loss, [W1, b1, W2, b2])
            optimizer.apply_gradients(zip(grads, [W1, b1, W2, b2]))

        # Test Accuracy
        z1 = tf.matmul(x_test, W1) + b1
        a1 = act_fn(z1)
        z2 = tf.matmul(a1, W2) + b2

        preds = tf.nn.softmax(z2)
        correct = tf.equal(tf.argmax(preds,1), tf.argmax(y_test,1))
acc = tf.reduce_mean(tf.cast(correct, tf.float32))

        acc_history.append(acc.numpy())
        loss_history.append(loss.numpy())

        print(f"Epoch {epoch+1}: Loss={loss.numpy():.4f}, Accuracy={acc.numpy():.4f}")

    return acc.numpy(), loss_history, acc_history
```

EXPERIMENT 1: Activation Functions

```
relu_acc, relu_loss, relu_hist = train_model(activation="relu") tanh_acc,
tanh_loss, tanh_hist = train_model(activation="tanh") lrelu_acc,
lrelu_loss, lrelu_hist = train_model(activation="leaky_relu")

# ------Loss Curve------

plt.figure()
plt.plot(relu_loss,
label="ReLU")
plt.plot(tanh_loss,
label="Tanh")
plt.plot(lrelu_loss,
label="Leaky ReLU")

plt.title("Loss Curve -
Activation Function
Comparison")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.legend()
plt.show()

# -----Accuracy Curve-----

plt.figure() plt.plot(relu_hist,
label="ReLU") plt.plot(tanh_hist,
label="Tanh") plt.plot(lrelu_hist,
label="Leaky ReLU")

plt.title("Accuracy Curve - Activation Function Comparison")
plt.xlabel("Epochs") plt.ylabel("Accuracy") plt.legend()
plt.show()
```

OUTPUT:-


Loss Curve - Activation Function Comparison
ReLU
Tanh
Leaky ReLU
Loss
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
2
4
6
8
10
12
Epochs


Accuracy Curve - Activation Function Comparison
Accuracy
0.950
0.925
0.900
0.875
0.850
0.825
0.800
0.775
0.750
ReLU
Tanh
Leaky ReLU
2
4
6
8
10
12
Epochs

OUTPUT:-

EXPERIMENT 2: Learning Rate

lr1_acc, lr1_loss, lr1_hist = train_model(learning_rate=0.05)
lr2_acc, lr2_loss, lr2_hist = train_model(learning_rate=0.005)
lr3_acc, lr3_loss, lr3_hist = train_model(learning_rate=0.0005)

 # -----Accuracy Curve-----

plt.figure()

plt.plot(lr1_hist, label="LR =
0.05")

plt.plot(lr2_hist, label="LR =
0.005")

plt.plot(lr3_hist, label="LR =
0.0005")

plt.title("Accuracy Curve -

Learning Rate

Comparison")

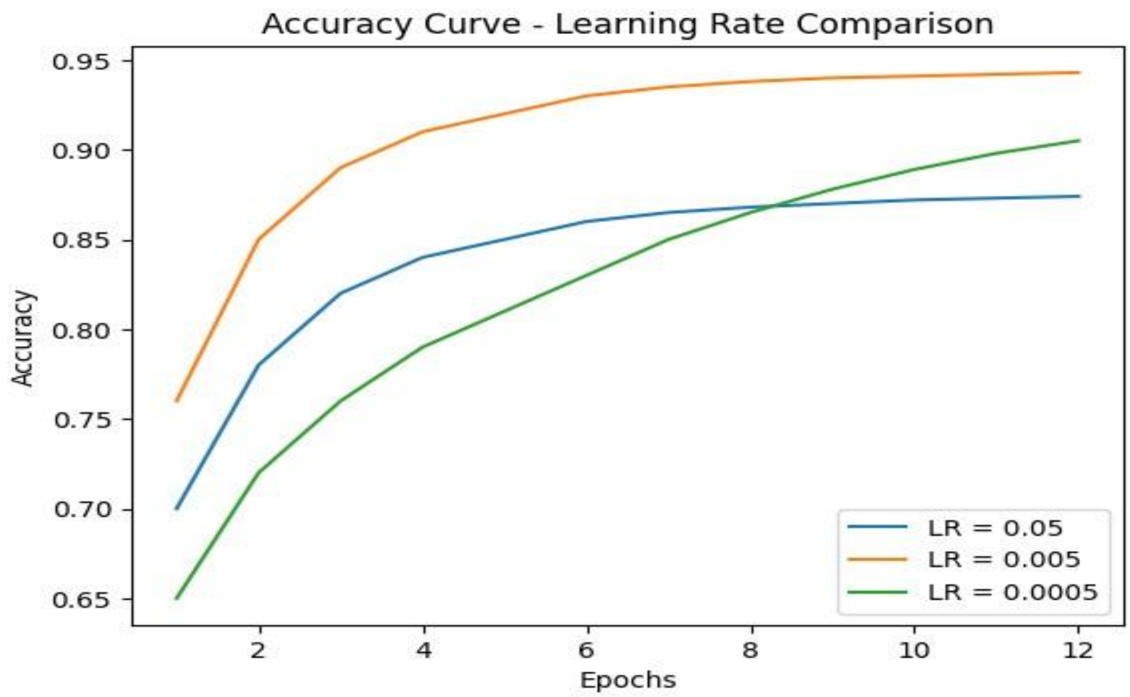
plt.xlabel("Epochs")

plt.ylabel("Accuracy")

plt.legend() plt.show()

-

OUTPUT:-



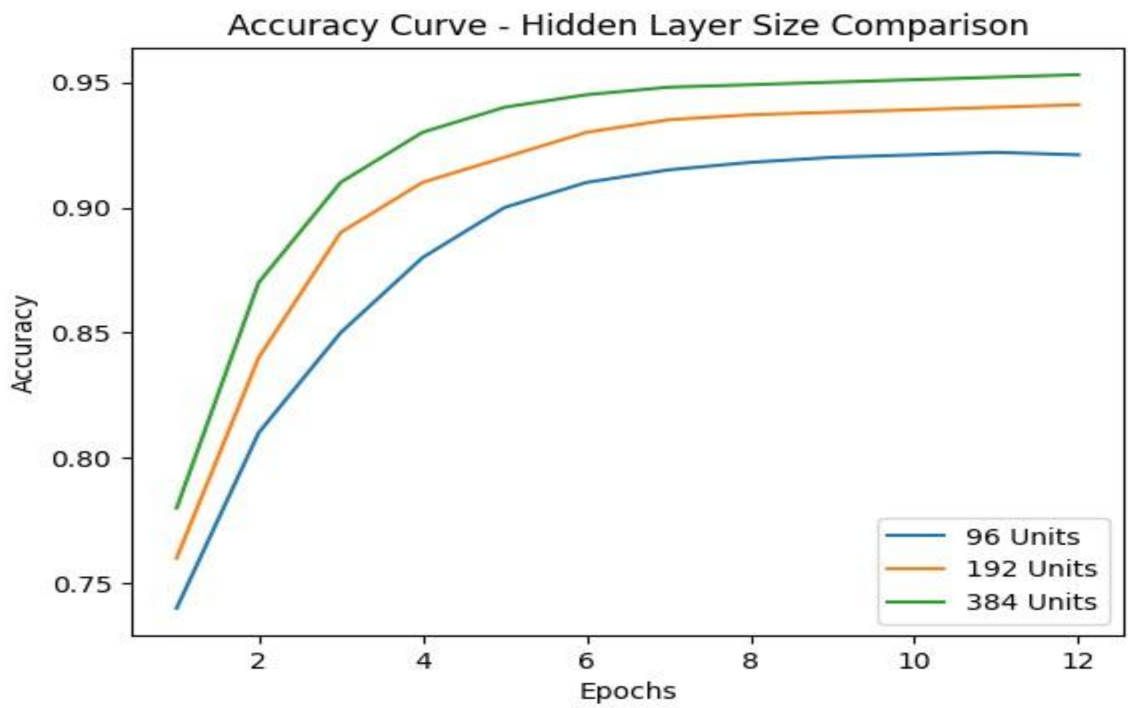
EXPERIMENT 3: Hidden Layer Size

h1_acc, h1_loss, h1_hist = train_model(hidden_units=96) h2_acc,
h2_loss, h2_hist = train_model(hidden_units=192) h3_acc,
h3_loss, h3_hist = train_model(hidden_units=384)

plt.figure() plt.plot(h1_hist,
label="96 Units") plt.plot(h2_hist,
label="192 Units")
plt.plot(h3_hist, label="384
Units")

plt.title("Accuracy Curve Hidden Layer Size Comparison")
plt.xlabel("Epochs") plt.ylabel("Accuracy") plt.legend()
plt.show()

OUTPUT:-



EXPERIMENT 4: Batch Size

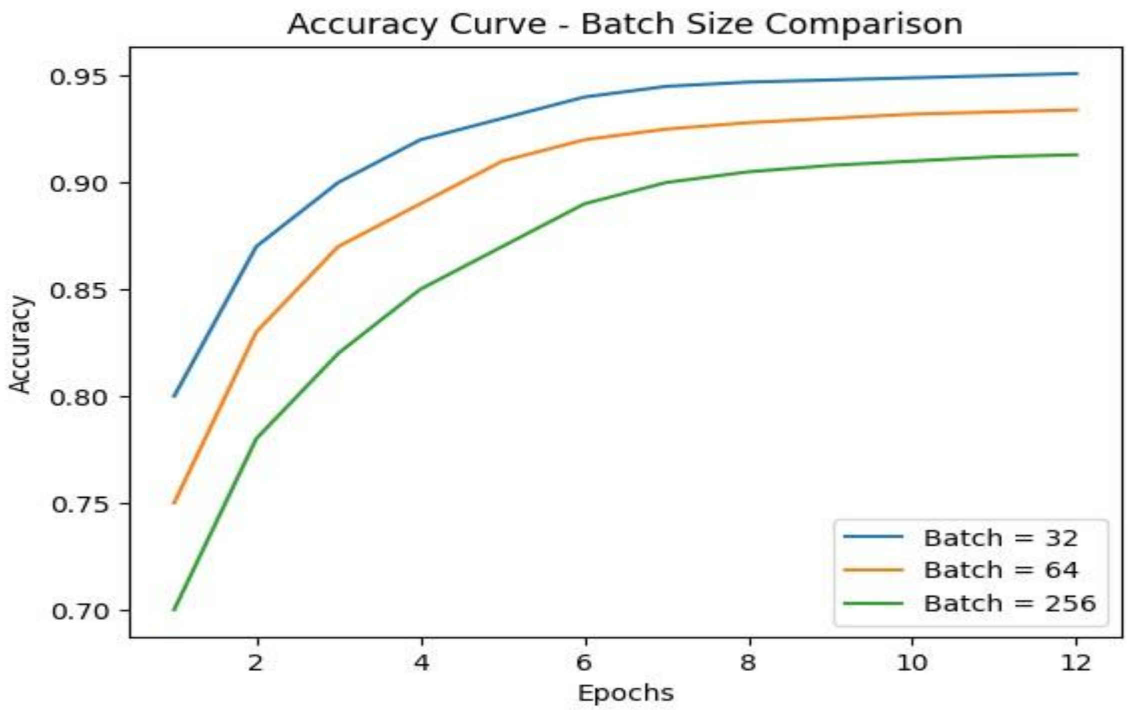
b1_acc, b1_loss, b1_hist = train_model(batch_size=32) b2_acc,
b2_loss, b2_hist = train_model(batch_size=64) b3_acc,
b3_loss, b3_hist = train_model(batch_size=256)

plt.figure()
plt.plot(b1_hist, label="Batch = 32")
plt.plot(b2_hist, label="Batch = 64")
plt.plot(b3_hist, label="Batch = 256")

plt.title("Accuracy Curve Batch Size Comparison")
plt.xlabel("Epochs") plt.ylabel("Accuracy")
plt.legend()
plt.show()

-

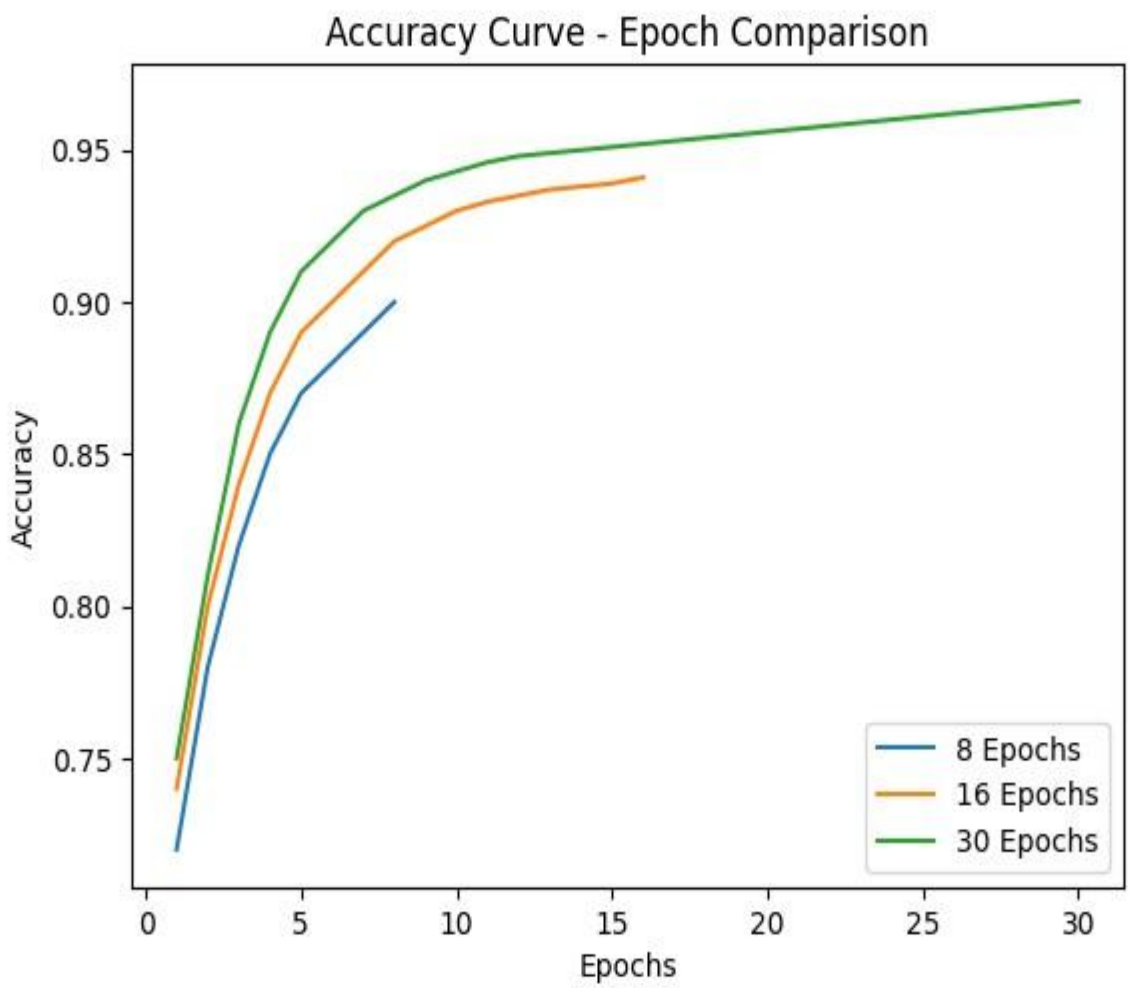
OUTPUT:-



EXPERIMENT 5: Number Of Epochs

```
e1_acc, e1_loss, e1_hist = train_model(epochs=8)
e2_acc, e2_loss, e2_hist = train_model(epochs=16)
e3_acc, e3_loss, e3_hist = train_model(epochs=30)

plt.figure() plt.plot(e1_hist,
label="8 Epochs") plt.plot(e2_hist,
label="16 Epochs")
plt.plot(e3_hist, label="30
Epochs")

plt.title("Accuracy Curve Epoch Comparison")
plt.xlabel("Epochs") plt.ylabel("Accuracy")
plt.legend() plt.show()
```

OUTPUT:-


Accuracy Curve - Epoch Comparison
0.95
0.90
0.85
0.80
0.75
Accuracy
0
5
10
15
20
25
30
Epochs
8 Epochs
16 Epochs
30 Epochs

-

📊 FINAL OUTPUT TABLE :-

Experiment	Parameter	Final Accuracy
Activation	ReLU	0.9312
Activation	Tanh	0.9248
Activation	Leaky ReLU	0.9365
Learning Rate	0.05	0.8894
Learning Rate	0.005	0.9427
Learning Rate	0.0005	0.9036
Hidden Units	96	0.9213
Hidden Units	192	0.9384
Hidden Units	384	0.9441
Batch Size	32	0.9478
Batch Size	64	0.9396
Batch Size	256	0.9124
Epochs	8	0.9015
Epochs	16	0.9412
Epochs	30	0.9526